On a metaplectic Jacquet–Langlands transfer for division algebras of prime degree (book) — Revision Log for Draft 1

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Decisions (David Joyner, November 8, 2026)

1. Combine Parts I–IV and the companion paper on the rank-two wild place into one book, so that cross-references and citations work.

2. Freeze the separate papers: their current drafts (Part I Draft 11, Part II Draft 10, Part III Draft 10, Part IV Draft 6, companion Draft 8) stay in the repository as history, and the book is the working document from now on, with its own drafts and logs.

3. Title: *On a metaplectic Jacquet–Langlands transfer for division algebras of prime degree*. Authors on the title page: David Joyner, Claude (Anthropic), ChatGPT (OpenAI); each chapter's source and author order are listed in the Preface.

Structure

Front matter	Preface (sources, conversion, contributions, the five acknowledgments)
1	Introduction (new; setting, chapters, running example, open problems)
2	Literature: Part I's review, then the related-work sections of Part II, Part III and the companion paper
3	Notation and glossary: the shared glossary, then the notation sections of Parts I, II, III and IV
4–8	Parts I, II, III, IV and the companion paper
9	Summary of results, one section per chapter (Part IV: its abstract)
Appendix A	Tables and verification scripts of the companion paper
Bibliography	merged, 104 entries

Each chapter opens with the source's abstract ("Overview") and a line naming the source draft. Numbering is by chapter and section (Theorem 4.4.5). The layout follows David's choices for Part I: literature and notation at the front, summary at the end.

No mathematical change

No statement, proof, hypothesis or status label was changed. Counting environments in the five sources and in the book gives identical totals: 26 theorems, 50 lemmas, 40 propositions, 14 corollaries,

16 definitions, 66 remarks, 5 examples, 128 proofs, 124 equations, 6 align blocks, 56 enumerations, 7 tables, 4 diagrams. Text removed: the five titles, abstracts (moved to the chapter overviews), the `\input` of the glossary in Parts I–III (the glossary appears once, in Chapter 3), the lead-in paragraphs of the three per-Part "Notation and glossary" sections (replaced by one lead-in for Chapter 3), and the five bibliographies (merged). The acknowledgments moved to the Preface. The build has no errors, no undefined references or citations, and no multiply defined labels.

Conversion rules

1. **Labels.** Every label of a source is prefixed by its chapter tag (`I:`, `II:`, `III:`, `IV:`, `W:`); the shared glossary's labels (`gl:`) are unchanged. Four headings that had no label but are cited from other Parts were given one: Part I §1.1 and §1.4, Part III §2, Part IV §2.1.

2. **Cross-Part references.** Typed references such as "Part I, Lemma 2.2" were mapped, by number, to labels using the compiled label tables of the drafts they were written against: Parts II and III and the glossary cite Part I Draft 11; Part IV, the companion paper and the running-example note cite Part I Draft 8 (= Draft 10) numbering; Parts II–IV have one numbering each. Theorem-like items are rendered with `\cref`, which prints the item's actual type; sections with `\S\ref`. "Part X" without a number became "Chapter **??**". 43 references typed without a "Part" prefix (the comparison table, diagram and restatements (I1)–(I11) of Part II; section pointers in Part III's lead-ins; one in Part I; five in the companion paper's list of dependencies; one "Appendix B" in Part IV) were mapped by hand; each is listed in `reports/assemble_fixes.txt`.

3. **Name mismatches.** The companion paper cites "Part I, Definition 2.2" twice (in its section on the cover in the Kazhdan–Patterson model, now §8.11, and in the dependency list of its main theorem). In Part I's Draft 8 numbering, item 2.2 is Lemma 2.2 (scalar centrality); Definition 2.1 defines the local covers and §2.2 is the ambient cover. The book renders the reference as `\cref` of item 2.2, so it now reads "Lemma 4.2.2". David: please check which item was meant (probably Definition 2.1, now Definition 4.2.1).

4. **Citations.** Bracketed citations (Parts I, II and the companion paper) were converted to `\cite` only when every key in the bracket is a key of that paper's bibliography, so that labels such as [P] and commutators such as $[Q, Q]$ were left alone. Citations of the Parts themselves (`P1`, `JoynerI`, ..., `RankTwo0`) became chapter references.

5. **Macros.** Each chapter keeps its source's macros, redefined at the start of the chapter inside a group, so different definitions of the same name (for example `\GDb`, `\GL`) do not interact. Theorem environments are defined once.

Bibliography

One key per work. 31 works appeared under several keys and were merged: BD, Bad, Ber, Coh, DKV, FK, Fli80, GG18, HC70, J, JC, JL, KP1, KP2, Kaz84, Klo86a, Kot84, Kub67, Kub69, Li2, Li4, Moo, PRa88, Rog83, Sav88, Ser, Wal80, Wal91, Wei, Wei18, Z (sources listed in `reports/bib_report.json`). Two key collisions were separated: [BZ] is Bernstein–Zelevinsky 1976 (Russian Math. Surveys) in Part I but 1977 (Ann. ENS) in Part II, now [BZ76] and [BZ77]; [Gao] is Gao's Mp_4 preprint in Part I but his 2018 Amer. J. Math. paper in the companion paper, now [Gao12] and [Gao18]. Where entries differed, Part I's (checked) entry was used, except DKV

(the companion paper's entry has pages) and Cohen (Part III's has the journal data). Weil's *Basic Number Theory* is cited in the 2nd edition by Part I and the 3rd by the companion paper; the entry names both. Moore's IHÉS paper now has the correct pages 5–70 (the companion paper had 157–222). Henniart's paper, in Part I's bibliography but never cited, was dropped.

Known limitations of Draft 1

1. Each chapter keeps its own introduction, so the setting, the labels and the running example are repeated; prose still says "this paper" and, in places, "here cited as Draft 8". An editing pass should remove the repetition and update the wording.

2. Parts IV and the companion paper had not had the exposition pass (summary, related work, lead-ins) given to Parts I–III; their existing summary and literature sections were used.

3. The chapter titles are those of the source papers; Chapter 8's title is shortened.

Files

`book/mjl_book_draft1.tex` and `.pdf`; `book/chapters/*.tex` (generated); `book/tools/` (the scripts that generated them, with a README); `book/reports/` (conversion report, hand fixes, bibliography report, source bibliographies); this log. The frozen papers are unchanged. README updated.